KeePass 2.0

Cross-Platform Password Manager with Browser Integration

Forge Hackathon 2026 Submission

Field	Details
Project	KeePass 2.0 - Cross-Platform Password Manager
Hackathon	Forge Hackathon 2026
Date	August 11, 2026
Team	Vignesh Subramanian, Shailesh Kolap
Repository	github.com/Vigneshsub30/keepass2.0
Release	github.com/Vigneshsub30/keepass2.0/releases/tag/v1.0.0
Forge Project	SoftwareForge.ai
Demo Video	(Google Drive link - to be added)

Problem Statement

KeePass is a widely trusted, open-source password manager - but it has been locked to Windows and WinForms for over two decades. macOS users have had no native option, and there is no browser integration for auto-filling credentials. The codebase also had security gaps that needed addressing.

Our goal: Transform KeePass into a modern, cross-platform password manager with native macOS support and seamless browser integration - all in a single hackathon session, powered by AI-assisted development through Cursor and SoftwareForge.ai.

What We Built

Before (Original KeePass)

- Windows-only WinForms application
- No macOS or Linux GUI support
- No browser extension integration
- Security issues: global TLS bypass, unsigned plugins, no vault integrity checks

After (Our Hackathon Work)

- Cross-platform desktop app built with Avalonia UI (MVVM architecture), supporting macOS, Linux, and Windows (end-to-end tested on macOS ARM64)
- Browser extension integration via the KeePassXC-Browser protocol
- Full credential management: create, edit, delete, search entries
- Password generator with configurable strength options
- Import/Export support for KDBX and CSV formats

- Custom app icon and polished DMG installer
- Security hardening across the entire codebase

Architecture

```
Chrome/Firefox          Native Messaging         keepass-proxy
(KeePassXC-Browser) <-- stdin/stdout (JSON) --> (Console App)
                                                      |
                                               Unix Domain Socket
                                                      |
                                                      v
                                              KeePass Avalonia App
                                               +----------------+
                                               | BrowserSocket  |
                                               | Server         |
                                               +----------------+
                                               | NaCl Crypto    |
                                               | (X25519 +      |
                                               |  XSalsa20-     |
                                               |  Poly1305)     |
                                               +----------------+
                                               | Entry Manager  |
                                               | Password Gen   |
                                               | Import/Export  |
                                               +----------------+
                                                      |
                                               KeePassLib (.kdbx)
```

Key Technical Achievements

1. Cross-Platform UI with Avalonia (MVVM)

Replaced the Windows-only WinForms frontend with a cross-platform Avalonia-based desktop app following the MVVM pattern. Supports macOS, Linux, and Windows from a single codebase. End-to-end tested on macOS ARM64.

2. Browser Extension Integration (KeePassXC-Browser Protocol)

Implemented the full KeePassXC-Browser native messaging protocol, enabling the existing Chrome/Firefox extension to communicate with our app for credential auto-fill.

3. End-to-End Encrypted Communication (NaCl)

All browser-to-app communication is encrypted using NaCl crypto_box (X25519 key exchange + XSalsa20-Poly1305 authenticated encryption) via a native messaging proxy.

4. Full Credential Lifecycle

Complete CRUD operations for password entries with a configurable password generator and import/export support for KDBX and CSV formats.

5. Security Hardening

Removed global TLS certificate validation bypass, added plugin signature verification, implemented vault integrity checks, and added security review sign-off gates.

Development Stats

Metric	Value
Forge work orders completed	15
Architecture Decision Records	8 ADRs
Development time	Single day
AI-assisted development	Cursor + SoftwareForge.ai

Forge Journey

This project was managed entirely through SoftwareForge.ai, which orchestrated the development workflow:

- Work Order Generation - Forge analyzed the KeePass codebase and generated targeted work orders covering security fixes, architectural decisions, and feature development.
- ADR-Driven Architecture - 8 Architecture Decision Records were produced to document key design choices (KDBX format selection, plugin trust model, clipboard delivery, MVVM adoption).
- Security-First Approach - Forge prioritized security work orders: TLS fix, plugin verification, vault integrity, and release promotion gates.
- Feature Development - After the foundation was solid, Forge guided feature work: cross-platform UI, browser integration, entry management, and packaging.
- Continuous Validation - Each work order included acceptance criteria verified through testing and validation.

Technology Stack

Layer	Technology
Language	C# / .NET 8
Desktop UI	Avalonia UI (MVVM, cross-platform)
Database	KeePassLib (KDBX 4.x format)
Crypto	NaCl.Net (X25519 + XSalsa20-Poly1305)
Browser IPC	Native Messaging Protocol (stdin/stdout)
App IPC	Unix Domain Sockets
Packaging	DMG (macOS), ZIP (Windows), DEB (Linux)
AI Tools	Cursor IDE, SoftwareForge.ai

Downloads

All artifacts available at: github.com/Vigneshsub30/keepass2.0/releases/tag/v1.0.0

Platform	File
macOS (Apple Silicon)	KeePass2.0-v1.0.0-osx-arm64.dmg

macOS (Intel)	KeePass2.0-v1.0.0-osx-x64.dmg
Windows (x64)	KeePass2.0-v1.0.0-win-x64.zip
Linux (x64, .deb)	KeePass2.0-v1.0.0-linux-x64.deb
Linux (x64, tar.gz)	KeePass2.0-v1.0.0-linux-x64.tar.gz

All builds are self-contained - no .NET runtime required on the target machine.

How to Run

macOS

1. Download the DMG for your Mac (ARM64 for Apple Silicon, x64 for Intel)
2. Double-click to mount, drag KeePass to Applications
3. Right-click > Open on first launch to bypass Gatekeeper
4. Open or create a .kdbx database file

Windows

1. Download and extract KeePass2.0-v1.0.0-win-x64.zip
2. Run KeePass.Desktop.Avalonia.exe

Linux

1. For .deb: sudo dpkg -i KeePass2.0-v1.0.0-linux-x64.deb, then run keepass
2. For tar.gz: extract and run ./KeePass.Desktop.Avalonia

Browser Extension

1. Install the KeePassXC-Browser extension for Chrome
2. With the app running and a database open, click the extension icon
3. Click Connect to pair the extension with the app
4. Navigate to any login page for auto-fill

Built with Cursor and SoftwareForge.ai during the Forge Hackathon 2026 - August 11, 2026.